



Predictive Modelling of Hybrid Rocket Engine Hot-Fire Tests



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Introduction

Hybrid rocket engines offer a promising balance between safety, performance, and simplicity when compared with other types. While hot-fire tests remain essential to validate performance, they are often expensive and time-consuming.



To complement experimental efforts, a zero-dimensional (0D) simulation framework to model the motor's internal behaviour was developed and compared with data from the hybrid rocket motor from project INVICTUS 1.0, which was developed to compete at Euroc2024.

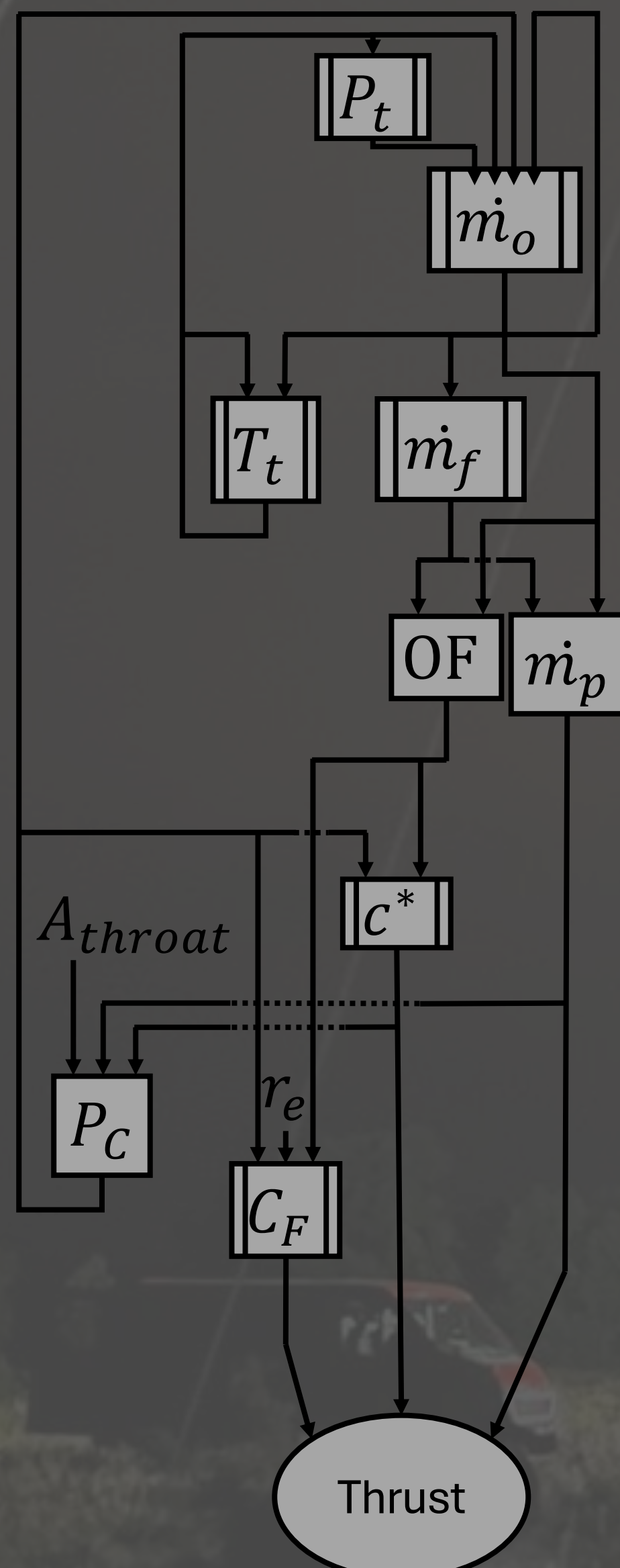
INVICTUS 1.0's hybrid motor was composed by 22 L oxidizer tank filled with nitrous oxide (N₂O) linked to a shower-head injector which is connected to the combustion chamber holding the fuel grain comprised of paraffin, ABS and some additives.



Materials & Methods

In order to measure the pressure of the oxidizer tank and combustion chamber during the hot-fire test, pressures transducers (RDP Group SPT-Z-G01500) were employed. The thrust was measured with 3 load cells (DAYSENSOR DYLY-103 500 kg).

To describe the hybrid rocket motor, several physical models were employed to describe the auto-pressurized oxidizer tank, multi phase N₂O mass flow at the injector, and the combustion and thrust kinetics at the combustion chamber. For thermodynamic data CoolProp [1] was utilized, and for combustion parameters from RocketCEA library [2], a NASA's CEA wrap, was utilized.



The tank pressure is the saturation pressure of the liquid at a given temperature, which is updated with the vaporization latent heat.

The oxidizer mass flow was determined with the non homogeneous non equilibrium model (NHNE):

$$\dot{m}_o = \frac{k}{1+k} * \dot{m}_{inc} + \frac{1}{1+k} * \dot{m}_{HEM}$$

The fuel mass flow was obtained by determining the regression rate:

$$\dot{m}_f = \frac{\Delta r * A_{comb} * \rho_{fuel}}{\Delta t} \quad \dot{r}(t) = a * G^n$$

After the mass flow are obtained, the OF ratio is established and the Thrust is determined, and the chamber pressures are updated:

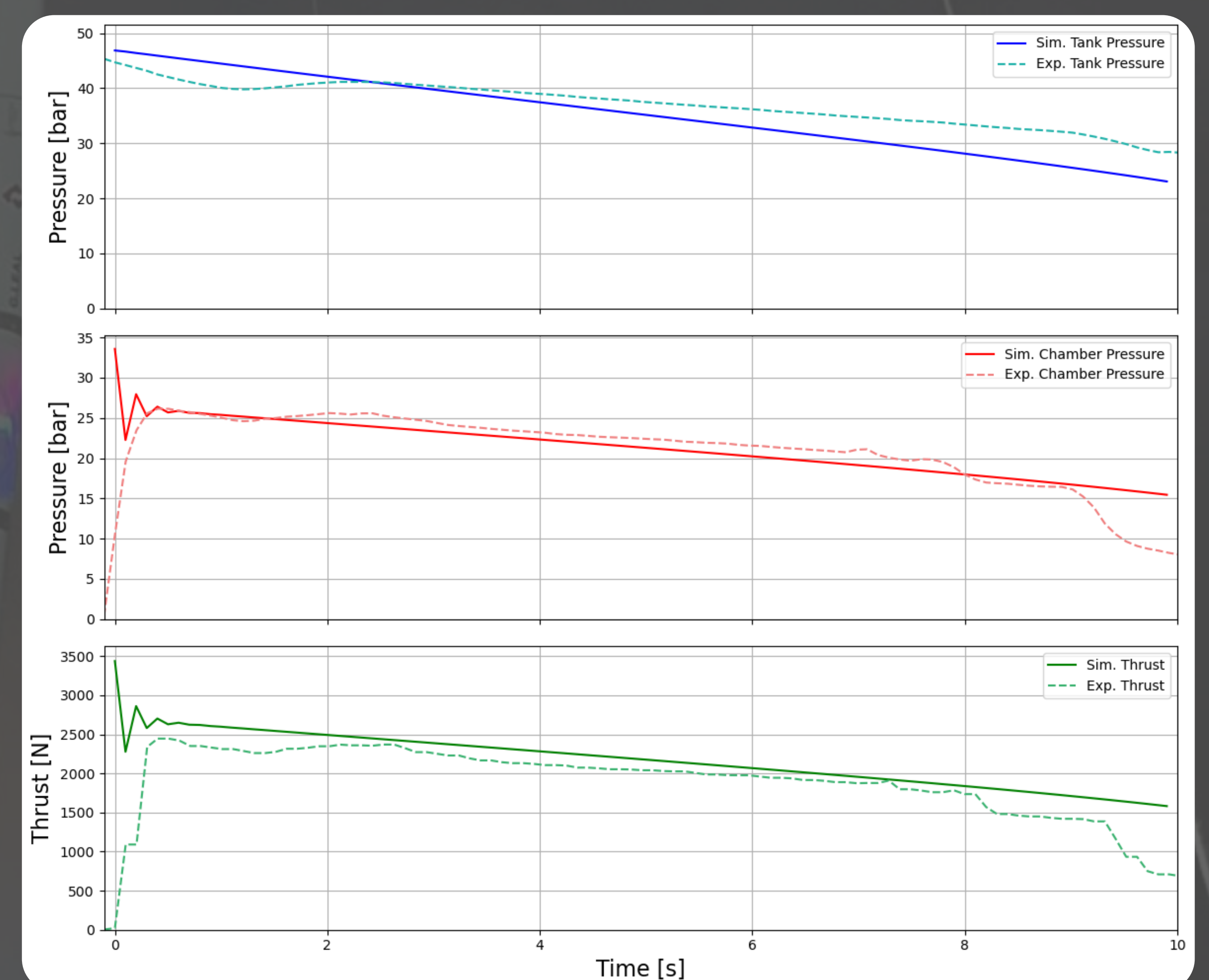
$$OF = \frac{\dot{m}_o}{\dot{m}_f} \quad P_c = \frac{(\dot{m}_o + \dot{m}_f) * c^*}{A_{throat}} * \xi$$

$$c^* = CEA(P_c, OF) \quad C_F = CEA(P_c, OF, eps)$$

$$F = (\dot{m}_o + \dot{m}_f) * c^* * C_F * \xi$$

Results & Discussion

In the graphs below, the tank pressure, combustion chamber pressure and thrust of both experimental and simulated data from the hot fire test are plotted.



The comparison between experimental and simulated data reveals good overall agreement in trends for tank pressure, chamber pressure, and thrust. However, the experimental results show greater fluctuation and a sharper drop near the end of the test, likely due to real-world factors such as flow instabilities.

These discrepancies highlight the need to refine the simulation by incorporating more realistic dynamics, such as fuel regression.

Conclusions

The comparison between experimental and simulated data from the hot fire test highlights the value and limitations of physical modelling in hybrid rocket propulsion, reaching the following conclusion:

- Simulated data captured overall trends in tank pressure, chamber pressure, and thrust.
- Experimental results showed fluctuations, and a sharper pressure drop, likely due to oxidizer depletion and instabilities.
- Thrust was overestimated in the simulation, suggesting simplified assumptions or input inaccuracies.
- Refining the model with fuel regression, two-phase flow, and injector dynamics is essential for better accuracy.

References

[1] – Bell, I. H. (2014). Pure and pseudo-pure fluid thermophysical property evaluation and the open-source thermophysical property library CoolProp. Industrial & Engineering Chemistry Research, 53(6), 2498–2508. <https://doi.org/10.1021/ie4033999>

[2] – Charlie, C. (2007). RocketCEA: A Python wrapper for NASA CEA code (Version 1.2.1) [Computer software]. Retrieved from <https://rocketcea.readthedocs.io/>

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